

LLM Chunking

VS Code Explorer:

- chunking - V4
 - .vscode
 - backend
 - .env
 - .env-original
 - app
 - data
 - test
 - .env
 - .env.example
 - requirements.txt
 - server.pid
 - server.stderr.log
 - server.stdout.log
 - frontend
 - app.js
 - index.css
 - index.html
 - llm-config.js
 - llm-config.test.mjs
 - .gitignore
 - Chunking_Playground_Implementation_Gui...
 - LLM_CONFIGURATION.md
 - README.md
 - start.ps1
 - Outline
 - Timeline

Chunking Playground - Visualizing RAG Document Segmentation

Buttons: Upload File, Web URL, Sample Web, Upload Document

Terminal Output:

```
INFO: 127.0.0.1:51797 - "GET /"
INFO: 127.0.0.1:51797 - "GET /"
INFO: 127.0.0.1:51798 - "GET /"
INFO: 127.0.0.1:51798 - "POST /"
INFO: 127.0.0.1:51798 - "GET /"
INFO: 127.0.0.1:51798 - "GET /"
INFO: 127.0.0.1:51797 - "POST /"
INFO: 127.0.0.1:51797 - "GET /"
INFO: 127.0.0.1:51801 - "GET /"
INFO: 127.0.0.1:51803 - "GET /"

python
Process ID (PID): 1660
Command line: ...powershell.exe -noexit -command 'try {
  "c:\Users\Admin\AppData\Local\Programs\Microsoft VS
  Code\7debd0e2a\resources\app\out\vs\workbench\contrib\terminal\common\scripts
  \shellIntegration.ps1" } catch {}'
Shell integration: injection failed to activate
Show Environment Contributions Show Details
```

Chat:

Implement LLM intelli... Then open <http://127.0.0.1:8000/>.

Worked for 26s >

The backend is responding now—I checked `/health` and received HTTP 200. The page was opened before startup finished.

Refresh <http://127.0.0.1:8000/> using the browser's reload button. Keep the backend terminal running.

Chunking on HR Leave Policy with fixed size character chunking

Chunking Playground - Document Preview

EMPLOYEE LEAVE POLICY AND GUIDELINES

1. Purpose and Scope

This policy outlines the guidelines and procedures governing various types of leave available to all full-time employees. The company believes that balance between professional and personal life is essential for productivity and well-being.

2. Annual Paid Time Off (PTO)

All full-time employees accumulate PTO at a rate of 1.67 days per month, resulting in a total of 20 days per calendar year.

- PTO requests must be submitted at least 5 business days in advance for approval by the department manager.
- A maximum of 5 unused PTO days may be carried over to the next calendar year. Any additional unused days will expire.

3. Sick Leave

Employees receive 10 paid sick days per year on January 1st. Sick leave is intended for personal illness, medical appointments, or caring for immediate family members.

- If sick leave exceeds 3 consecutive days, a certified medical note from a

Analysis

TOTAL CHUNKS: 4

AVG. CHUNK SIZE: 322.2 chars

EST. TOKENS: 322

TIME ELAPSED: 0.288 ms

Sample Library

- HR Leave Policy (Policy - 2.1 KB)
- Company FAQ (FAQ - 1.5 KB)
- RAG Research Abstract (Paper - 4.1 KB)
- Banking Loan Policy (Banking - 2.1 KB)

Recursive Character Chunking Educational Insights

Hierarchy Preservation Metrics

Metric	Value
Paragraph Preservation %	100%
Line Preservation %	100%
Word Breaks Count	0 / 50,000
Character Splits (Forced)	0 / 100,000

Separator Usage Analytics

Separator	Count
Paragraph (¶)	3
Line (↵)	0
Word (")	0
Char (")	0

Fixed vs Recursive Split Comparison

Metric	Fixed Chunking	Recursive Chunking
Paragraph Preservation	60%	100%
Line Preservation	82.4%	100%
Word Boundary Breaks	1	0
Sentence Boundary Breaks	2	0

Dynamic Hierarchy Preservation

Why is Recursive better than Fixed?

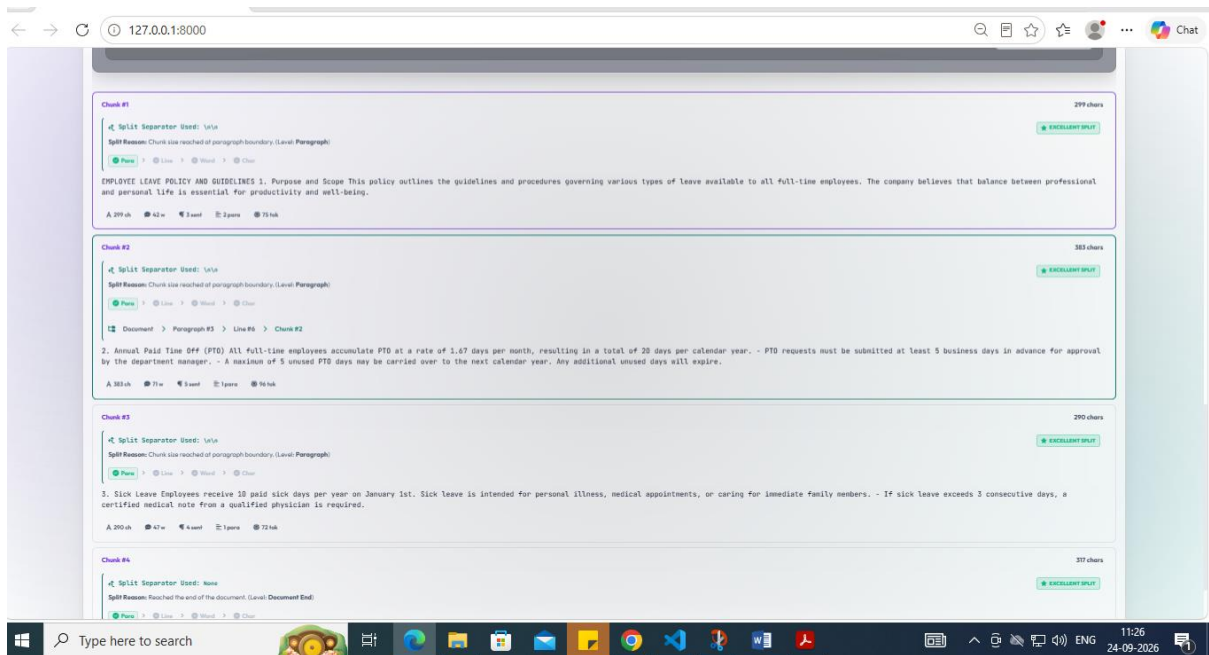
Fixed-size chunking slices text strictly by character length, which frequently splits words (e.g., `comput-er`) and cuts sentences in half. Recursive chunking uses a hierarchical fallback system: it attempts splits on paragraphs first, then lines, then word spaces. This ensures word boundaries are never broken and sentences remain complete inside chunks.

Why is Recursive STILL not Semantic?

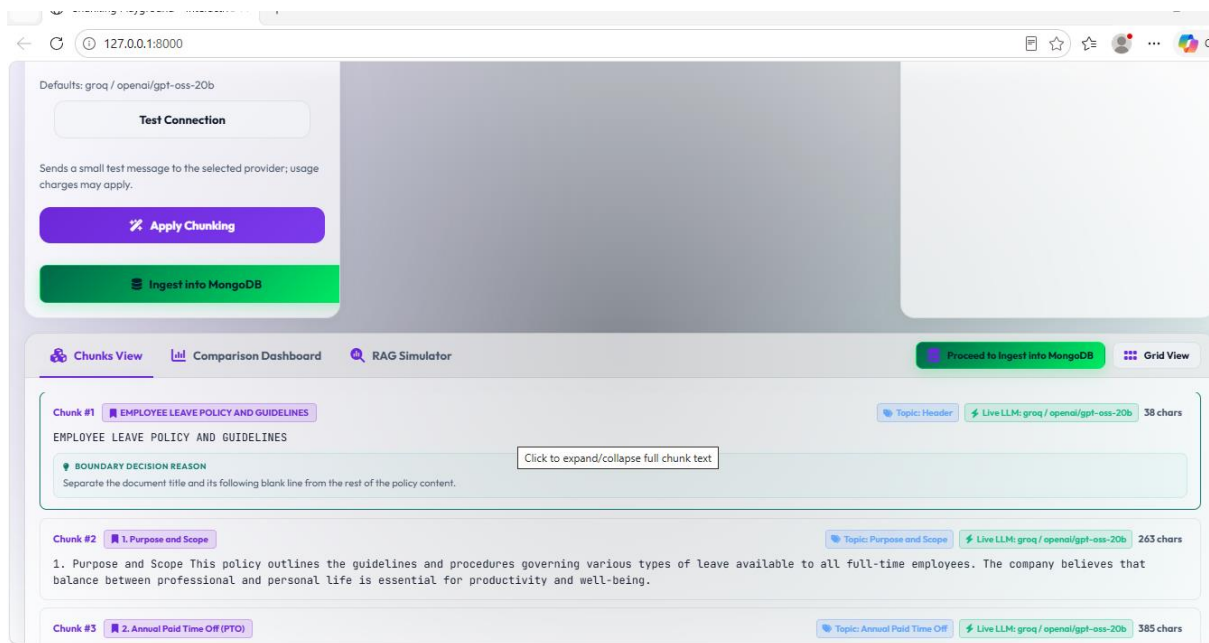
While recursive chunking preserves structural text segments (like paragraphs and lines), it does not analyze the semantic meaning or content transitions of sentences. If a single topic spans across paragraphs, recursive splitting might partition them into separate chunks purely based on size limits, whereas Semantic Chunking evaluates topic embedding similarities to group related paragraphs coherently.

Recursive Process Walking Simulator

Show Recursive Process



LLM Based Hr Policy chunking:



Temperature Settings

Chunking Playground – Interactive

127.0.0.1:8000



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Confirm

Options

https://api.groq.com/openai/v1/chat/completions

API Key

Leave blank to use backend key

Backend key configured (hidden). Leave the key field blank to use it.

Temperature (optional)

0.2

0.2 – Low variation

Temperature presets...

0 – Least variation

0.2 – Low variation

0.7 – More variation

1 – High variation

Use backend setting

Use provider default (omit temperature)

Sends a small test message to the selected provider; usage charges may apply.

 Apply Chunking

Case	Document and experiment	Observation
1. Basic comparison	Use an employee policy containing leave, attendance, benefits, and resignation sections. Run fixed-size, semantic, metadata-enhanced, and LLM chunking .	Do chunks contain complete policies? Are unrelated policies combined? Does fixed-size split a rule midway?
2. Topic changes without headings	Use several paragraphs discussing recruitment, then payroll, then remote work, without section titles. Compare semantic and LLM chunking.	Do boundaries follow changes in meaning despite missing headings?
3. Rules and exceptions	Use a leave policy where eligibility appears in one paragraph and exceptions appear in the next. Compare all four strategies.	Does each exception retain enough context to identify the rule it modifies?
4. Procedures and lists	Use an onboarding guide with numbered steps, required documents, and approval conditions.	Are related steps preserved together? Are list introductions separated from their items?
5. Metadata and context	Use policies for HR and IT with explicit department, version, and effective-date information. Compare metadata-enhanced and LLM chunking.	Which metadata does the app retain? Can a chunk be traced to its department and version? Topic labels alone are not equivalent to source metadata.
6. Temperature variation	Run the same document three times each at 0, 0.2, and 0.7 , using the same provider and model.	Compare chunk count, boundaries, and consistency. Even temperature 0 may produce variation.
7. Document size and limits	Start with a short policy, then progressively add sections. Compare LLM and fixed-size chunking.	Record processing time, failures, oversized chunks, and whether content is missing. Distinguish rate limits from input/output limits.
8. Retrieval usefulness	Use the app's RAG Simulator to ask questions whose answers span a rule and its exception. Compare the chunks retrieved for each strategy.	Does retrieval include the complete answer and relevant conditions? If generation is simulated, evaluate retrieval rather than claiming real answer-quality improvements.